

Karan Patel

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EDUCATION

The George Washington University

M.S. in Computer Science

Washington, DC

August 2024 – May 2026

SVKM's NMIMS

B.Tech in Computer Engineering, Honors in AI

Mumbai, India

August 2018 – May 2024

EXPERIENCE

TWODS Capital

Remote

Co-Founder & Software Engineer

September 2019 – Present

- Built and automated a financial-education platform that generated 104K+ organic search clicks and 658K+ search impressions at a 15.9% click-through rate over a 12-month window, growing organic reach with zero paid acquisition.
- Built QuantEra, an equity-research platform that computes fundamental and technical analysis deterministically and constrains a language model to narrating computed results, behind a fail-closed guardrail that discards any untraceable number or source.
- Validated the QuantEra data pipeline against hand-entered SEC 10-K figures for 8 sector-diverse large-caps, reaching 95.8% field-level accuracy with a period-aware harness that refuses to pass on unverified ground truth.

George Washington University

Washington, DC

Technical Assistant, Software Platform Systems

July 2025 – May 2026

- Designed and developed a high-performance ticket-routing pipeline in Python, implementing hash map and queue-based data structures and algorithms to classify and route requests across globally distributed media systems.
- Built low-latency REST API services integrating the Freshdesk and Kaltura APIs, handling high-throughput event streams across multiple data centers with automated health checks and debugging.
- Collaborated with product and engineering teams to influence system design and built real-time monitoring dashboards supporting high availability across large-scale infrastructure.

Vipul Enterprise

Mumbai, India

Software Development Engineer

June 2023 – July 2024

- Built Python and FastAPI inventory and order APIs for 10K SKUs across 5 warehouses, implementing stock reservations and row-level locking to eliminate overselling, raising stock accuracy from 92% to 96% and reducing related support tickets by 80%.
- Optimized PostgreSQL stock-reporting queries with composite indexes and N+1 query elimination, cutting p95 latency 75% from 1.4s to 350ms, and automated low-stock alerts with Celery and Redis to remove 2 hours/day of manual monitoring.
- Analyzed inventory and sales patterns to identify slow-moving stock, stockout risk, and reorder points, contributing to a ~15% profit improvement across business operations.

Junior Software Engineer

August 2022 – June 2023

- Built REST APIs in Node.js and Python for stock monitoring, supplier tracking, order lifecycle management, and reporting across purchasing, sales, and warehouse workflows, with React interfaces for real-time stock-level tracking.

PROJECTS

CodeKraft | Python, CodeBERT, FastAPI, React, GPU Inference

[GitHub](#)

- Fine-tuned a CodeBERT transformer for Python error classification, achieving CodeBLEU 52.3% and token-level F1 75.4% on benchmark evaluation.
- Engineered a 4-layer analysis pipeline (static AST analysis, transformer classification, rule engine, async enrichment) serving feedback in sub-300ms on a Tesla T4.
- Built a FastAPI inference service with a React frontend, deployed serverless on Vercel with GPU-optimized model serving.

BNPL Marketplace | Python, FastAPI, MySQL, Redis, Docker, Kubernetes, GitHub Actions

[GitHub](#)

- Built a high-performance distributed backend with versioned REST endpoints and an async job-state machine over MySQL, delivering sub-4ms P95 search.
- Engineered a Redis-backed caching and rate-limiting layer with relevance-ranked search and sub-1ms autocomplete across 600 merchant records.
- Deployed containerized services to Kubernetes with horizontal pod autoscaling, health probes, and ingress, with a GitHub Actions CI pipeline validating 53 tests at 90% coverage.

LEADERSHIP & HONORS

- Recognized at the H2AI Georgetown Hackathon for MindMate, a full-stack mental-health AI platform (React, Node.js, MongoDB, Gemini).
- Selected as Perplexity Campus Partner at George Washington University.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++

Backend: REST APIs, FastAPI, Node.js, Express, React, Microservices, OOD, System Design, Distributed Systems

Infrastructure: MySQL, PostgreSQL, MongoDB, Redis, Docker, Kubernetes, AWS (EKS/ECR), CI/CD, SDLC, Agile